Reproducing ViDoRe v3 with open-weights models

Does adding image retrieval help a strong text-based RAG?

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Why hybrid RAG?

RAG pipelines over visually rich PDFs face a basic tension: the answer may live in **text** (paragraphs, tables) *or* in **visual elements** (charts, figures, handwriting) that text extractors drop or flatten.

Two retrieval paradigms have emerged:

- **Textual retrieval**: parser → markdown → dense embedding. Precise on text; blind to figures.
- **Visual retrieval**: page-image encoder (ColPali / ColEmbed). Sees layout and figures; less precise on fine-grained text.

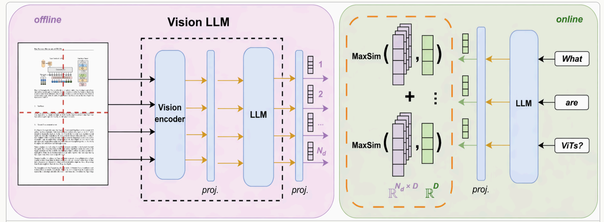
> **Hybrid** concatenates the top-k of both streams and feeds them to a multimodal generator. The intuition: the two streams capture *complementary* evidence. Does that hold in practice?

Visual retrieval

Visual retrieval: each PDF page is encoded *as pixels* by a vision-language model, once, offline. No OCR, no markdown, no chunking. A page is its own atom. At query time, pages are ranked by image-level similarity to the query.

ColPali (Faysse et al., 2025)

Late interaction: encodes each page into one embedding per image patch, and each query into one embedding per token. Scoring uses **MaxSim**: for every query token, find its closest page patch; sum those best-match scores across all tokens. A query asking "what is the Q3 revenue?" can match "Q3" to the right table column and "revenue" to the right row, something impossible when the whole page is collapsed into a single vector.



Offline: a vision-LLM (PaliGemma, 3B) encodes each page into patch embeddings. **Online**: the query is tokenized through the same LLM; score = **MaxSim**.

ColEmbed (Xu et al., 2025)

Same late-interaction architecture, two upgrades: (i) stronger backbone, **Llama-Nemotron** (NVIDIA's document-focused VLM family); (ii) broader synthetic query–page training data with hard-negative mining.

The ViDoRe v3 benchmark

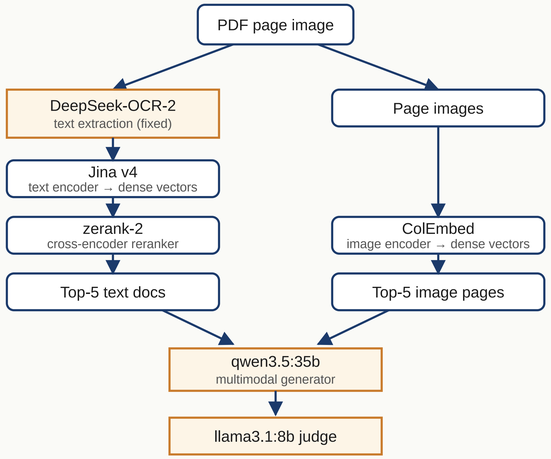
Loison et al. (2026): 10 corpora, **26 000 pages**, **3 099 human-verified queries** translated into 6 languages. We evaluate on 5 public subsets (3 English + 2 French), totalling **10 673 pages** and **2 132 queries**.

What makes it hard

- **Visually rich documents**: annual reports, scientific books, pharma regulations. Tables, charts, figures carry real evidence.
- **7 query types** of very different difficulty: from open-ended synthesis to single-number lookup.
- **Cross-lingual**: queries in 6 languages paired with english or french source documents.
- **Decoupled evaluation**: the paper reports retrieval quality (NDCG@10) and end-to-end answer accuracy (pass@1) separately, and shows the two disagree: the best retriever is not always the best generator.

Method

> **Question**: with the text stream fixed at DeepSeek-OCR-2 + Jina v4 + zerank-2, does adding a visual retrieval stream (ColEmbed) improve *end-to-end* answer accuracy?



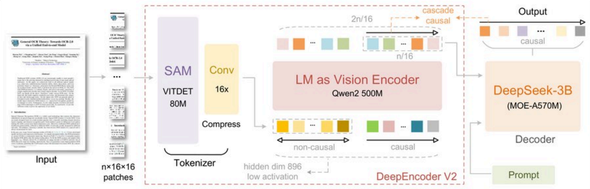
Model choices

DEEPSEEK-OCR-2	open-weights document OCR with layout-preserving markdown (HTML tables, LaTeX); designed for faithful reconstruction
QWEN3.5:35B	multimodal (text + image) generator, Q4-quantized MoE, fits on one 48 GB GPU; 262 k context
LLAMA3.1:8B	judge in a *different* model family (Meta vs Alibaba) to reduce self-preference bias; officially multilingual (EN + FR covered)

All components are open-weights and run on a single GPU. We reproduce the paper's findings without the closed Gemini 3 Pro / GPT-5.2 stack.

DeepSeek-OCR-2

Instead of consuming visual tokens from top-left to bottom-right, learnable **causal flow queries** re-order the tokens to a more logical reading path.



Results

Retrieval (NDCG@10)

Subset	colembed (image)	jina + zerank-2 (text)
computer_science (EN)	78.0	**82.4**
finance_en (EN)	**68.6**	65.7
pharmaceuticals (EN)	**67.2**	65.1
physics (FR)	**47.8**	46.6
finance_fr (FR)	47.6	**48.1**
avg	**61.8**	**61.6**

End-to-end (pass@1)

Subset	closed-book	colembed (image)	jina + zerank-2 (text)	hybrid
computer_science	91.6	93.0	**95.3**	**95.3**
finance_en	56.0	74.8	79.0	**81.6**
pharmaceuticals	68.4	83.2	88.5	**89.3**
physics	87.8	83.1	**90.1**	89.7
finance_fr	40.6	70.3	**80.3**	77.2
avg	68.9	80.9	86.6	**86.6**

Where does retrieval actually help?

pass@1 (%) per query type:

query type	closed	image	text	hybrid	example query
numerical	36.4	70.4	70.7	**74.1**	"What is the difference in the total number of FAERS reports of 2015 and 2018?"
extractive	54.3	77.4	**83.1**	82.6	"What was Bank of America's net income in 2024?"
multi-hop	68.6	84.7	84.7	**91.5**	"What is a while loop, and how is it used to create a nested loop?"
enumerative	60.3	75.8	**83.5**	80.9	"What are the five phases of relational database design (RDD)?"
boolean	72.1	85.0	**93.2**	**87.8**	"Did Bank of America's efficiency ratio improve in 2024?"
compare-contrast	71.9	80.8	84.6	**85.3**	"How do heuristics differ from process patterns?"
open-ended	82.8	82.6	**91.9**	91.5	"How does REMS help in managing the opioid crisis?"

Does image retrieval help?

> **Sometimes. It depends on the query.**

When images help

- **Multi-hop** (+6.8 over text): evidence spans multiple pages; the two streams retrieve partly *different* pages, giving the generator broader coverage.
- **Numerical** (+3.4): visual table layout (rows, columns, cell alignment) preserves structure that OCR partly loses.

When images don't help (or hurt)

- **Boolean** (−5.4): text already retrieves the fact; extra image pages dilute the prompt and push the generator toward hedged, less decisive answers.
- **Extractive / enumerative / compare-contrast / open-ended**: flat (±1 pt). The answer lives in a text span the text retriever already finds.

Takeaways

- **The 86.6% tie hides structured query-type variation.** Hybrid wins on multi-hop (+6.8 pts) and numerical (+3.4); text wins on boolean (+5.4). The right choice depends on your workload's query mix.
- **A strong text pipeline is already very competitive.** DeepSeek + Jina + zerank-2 matches hybrid at half the retrieval cost. Images add value only for multi-hop and numerical workloads.
- **Retrieval matters most where the LLM's prior knowledge runs out.** Gains over closed-book: +37.7 pts on numerical, +28.3 on extractive, only +8.7 on open-ended.
- **The open-source generator is likely the bottleneck.** ViDoRe v3 reports +2.6 pts hybrid gain with Gemini 3 Pro. Our null result reflects a weaker vision stack, not uninformative images.

Limitations

LLM choices

- **Generator (qwen3.5:35b)**: its vision stack is weaker than frontier closed models. The paper reports +2.6 pts hybrid gain with Gemini 3 Pro; our null result is likely *generator-limited*, not a property of the image stream.
- **Judge (llama3.1:8b)**: LLM-as-judge has a known **verbosity bias**. Hybrid answers tend to be longer, so some gains could be style-driven rather than content-driven.

Benchmark and evaluation design

- **Binary pass@1**: a near-miss and a complete miss are treated identically. No partial credit.
- **No human spot-check**: all correctness judgments come from the LLM judge with no human validation.
- **Fixed fusion strategy**: always top-5 text + top-5 image with no deduplication. The generator can receive the same page twice, wasting context.
- **Domain-specific corpora**: CS textbooks, finance reports, pharma regulations. Results may not transfer to general-purpose RAG.

Future work

- Rerun with stronger multimodal generators (**Gemini 3 Pro**, Qwen3-VL-235B) to isolate the generator-capability effect.
- Apply a controlled-verbosity prompt and swap the judge to rule out verbosity bias.
- Test adaptive fusion strategies (weighted combination, deduplication, query-type routing).
- Figure captioning to close the figure-placeholder blind spot in the text stream.

Text parser: DeepSeek-OCR-2 · Text retriever: Jina v4 · Reranker: zerank-2 · Visual retriever: ColEmbed (Llama-Nemotron 3B v2) · Generator: qwen3.5:35b · Judge: llama3.1:8b · Benchmark: ViDoRe v3 (Loison et al., 2026)